

Prediction of Antimicrobial Resistance from Whole-Genome Sequences Using Neural Networks

First Author, Second Author, Third Author, and Fourth Author

Placeholder: summarize the problem, dataset, shallow neural network approach, deep neural network approach, final metrics, and main conclusion in 150–250 words. The abstract should be self-contained and written as one paragraph.

Index Terms—Antimicrobial resistance, deep learning, k-mers, neural networks, whole-genome sequencing.

I. PROJECT GOAL AND SCOPE

PLACEHOLDER: state the project goal in a few executive sentences. Include the biological problem, the prediction target, and the intended scope of the shallow and deep neural network approaches.

II. LITERATURE REVIEW

Placeholder: define antimicrobial resistance, whole-genome sequencing, k-mers, binary classification, shallow neural networks, recurrent neural networks, attention, and evaluation metrics. Cite the most relevant background sources, for example neural network textbooks and AMR/WGS references [1], [2].

III. CASE STUDY

Placeholder: describe the specific paper or documented advance that motivates the project. Explain why it is relevant, what part of the approach it supports, and what limitations remain for this project.

IV. TRAINING DATA SET

Placeholder: describe the source of the data, organisms included, antimicrobial susceptibility labels, number of genomes, number of genome-antibiotic records, class balance, and train/validation/test split. Clarify whether the split is stratified by genome ID.

V. DATA PREPROCESSING AND EXPLORATORY DATA ANALYSIS

Placeholder: describe data cleaning, label normalization, FASTA processing, k-mer extraction, feature normalization, and exploratory analysis. Include the key plots and tables needed to summarize class imbalance, species distribution, antibiotic distribution, missingness, and relevant feature properties.

VI. PROPOSED NEURAL NETWORK-BASED APPROACHES

A. Shallow Neural Network-Based Approach

Placeholder: describe the shallow baseline model. Explain feature engineering, feature selection or extraction, the final input vector, architecture, activation functions, regularization, optimizer, loss function, and threshold calibration.

Neural Networks, 2026-I. Term project final report. Replace this note with team ID, affiliations, and submission metadata.

Placeholder figure: EDA summary, class balance, or data pipeline.

Fig. 1. Placeholder for an exploratory data analysis figure. Replace with a real figure from the project.

TABLE I
PLACEHOLDER FOR FINAL MODEL COMPARISON

Model	F1	Recall	Precision	AUC
Shallow NN	–	–	–	–
Deep NN	–	–	–	–

B. Deep Neural Network-Based Approach

Placeholder: describe the deep model. Explain the selected architecture, input representation, antibiotic embedding, recurrent/attention/set encoder components, classifier head, regularization, optimizer, loss function, and why this architecture was chosen over alternatives.

C. Learning, Validation, and Testing Protocol

Placeholder: explain the training loop, validation strategy, early stopping, checkpointing criterion, random seed, class imbalance handling, and final test evaluation protocol.

VII. PERFORMANCE EVALUATION AND FINAL RESULTS

Placeholder: report final metrics for the shallow and deep models. Include F1, recall, precision, AUC-ROC, confusion matrix, and any threshold used. Diagnose overfitting or underfitting using training and validation curves.

VIII. ETHICAL CONSIDERATIONS AND QUESTIONS

Placeholder: discuss clinical risk, false negatives, false positives, dataset bias, generalization limits, transparency, reproducibility, privacy, and responsible communication of model limitations.

IX. WEB LINKS TO SOURCE CODE AND EXPLANATORY VIDEO

A. Source Code

Placeholder: provide the GitHub repository or equivalent URL. Briefly explain how the repository is organized and where the shallow and deep model implementations can be found.

TABLE II
PLACEHOLDER FOR TEAM CONTRIBUTIONS

Team member	Role	Activities/contributions
Member 1	–	–
Member 2	–	–
Member 3	–	–
Member 4	–	–

B. Explanatory Video

Placeholder: provide the URL to the explanatory video and summarize what the video covers.

X. CONTRIBUTIONS OF THE TEAM MEMBERS

Placeholder: describe roles, responsibilities, and concrete contributions. Replace Table II with the final team information.

XI. CONCLUSIONS AND FUTURE WORK

Placeholder: summarize the main findings, compare shallow and deep approaches, identify limitations, and propose future work such as biological annotation, calibration analysis, broader validation, or improved representations.

XII. USE OF ARTIFICIAL INTELLIGENCE TOOLS

Placeholder: disclose whether AI tools were used to draft, review, debug, summarize, or generate code. State which tools were used, what parts they supported, and how the team verified the outputs.

ACKNOWLEDGMENT

Placeholder: optional. Include course, instructor, institutional, or data-source acknowledgments if appropriate.

REFERENCES

- [1] S. Haykin, *Neural Networks and Learning Machines*, 3rd ed. Pearson, 2009.
- [2] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.